

COMS4995W30

Applied Machine Learning

Dr. Spencer W. Luo

Columbia University | Fall 2026

About this Course



COMS4995W30 - Applied Machine Learning

- Schedule: Thursday 4:10pm - 6:40pm, Fall 2026
- Location: 717 Hamilton Hall (Morningside Campus)
- Credits: 3.0

Instructor



Dr. Spencer Luo

- Current Role:
 - Principal Research Scientist @Google DeepMind
- Background:
 - 🎓 Ph.D. in Artificial Intelligence, Carnegie Mellon University ☕ 🧋
 - 🚗 Self-Driving Startup 😵 🎉
 - 🤖 OpenAI 😄 😓 🌀
 - 🏠 Google 🚀

Our GREAT TAs



- Zoga Duka (zd2377@)
- Prajwal Raghunath (pr2789@)
- Byeolah Kwon (bk2833@)
- Kuan-Ting Chen (kc3953@)

Course Setup



<https://columbia-coms4995.github.io/aml-fall2026/>

- [Syllabus](#) - Please read it carefully
- [Ed Discussion](#)

Lectures: Weekly applied ML topics

- 3 assignments (Code + Report) - 60%
 - 1 midterm exam - 20%
 - 1 final exam - 20%
-
- Office Hours: (To be announced on Ed soon)

Enrollment



Fully handled by DSI student affairs now, please talk to them **directly**

POC: Robert Kramer (rk3281@columbia.edu)

Overview



- Foundations of Applied ML
 - ML workflow, in production, and case studies
 - Data preparation, cleaning, and feature engineering
- Classical ML Methods
 - Generative vs. discriminative models
 - Evaluation metrics, bias–variance tradeoff
 - Tree-based models and ensemble methods

Overview



- Deep Learning

- Neural networks fundamentals (MLP, backprop, activation)
- Transformers (Attention, BERT, ChatGPT, Gemini)

- Large Language Model

- Pre-training & supervised fine-tuning
- Retrieval-augmented generation
- Agentic workflows (Thinking model, LangChain, Tool integration)

This Week in AI - Race to Zero & Agentic Frontier



The Landscape (Fall 2026):

- Cost War: Inference costs dropped >95% since Jan 2025; raw token pricing is commoditized.
- Frontier Shift: Competition moved from parameter scale to test-time compute, vibe coding, and autonomous agents (Gemini, Claude, DeepSeek).

AML Takeaway:

- Model performance is converging.
- System efficiency (Latency, Serving Cost, Reliability) is the real moat.



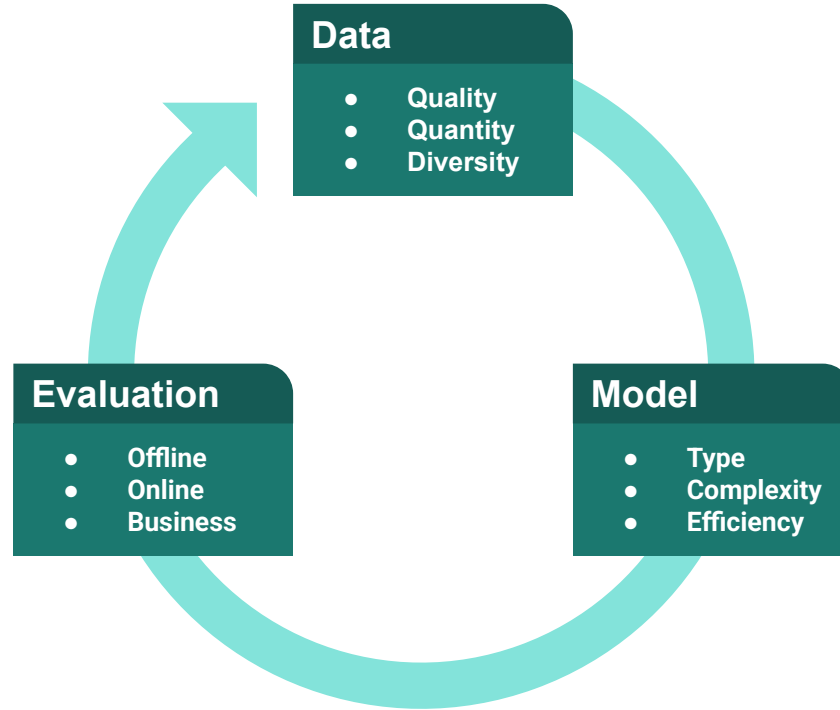
Introduction to AML



What is Machine Learning (ML)?

- Learn patterns from **data** 
- Make **predictions** on new data 
- **Generalize** beyond training 

The Applied ML Life Cycle





Types of Learning

Supervised Learning

- [Data] labeled inputs → outputs
- [Goal] predict correct labels on new data
- [Ex]  Spam /  Diagnosis /  Ad click

Unsupervised Learning

- [Data] unlabeled, only features
- [Goal] discover hidden structure
- [Ex]  Anomaly detection /  Image segmentation

Types of Learning



Reinforcement Learning

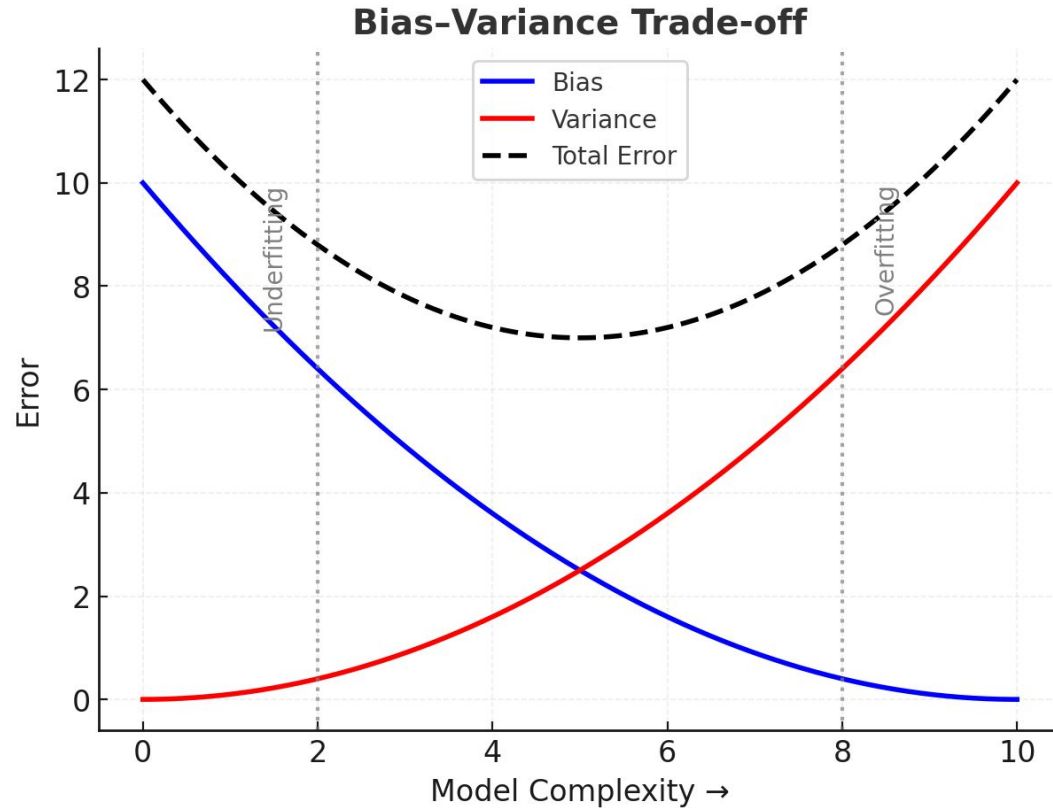
- [Data] interaction with environment
- [Goal] maximize expected cumulative reward
- [Ex]  Self-driving car /  AlphaGo

Trade-offs Discussions - Key in Applied Science



- Bias vs. Variance
 - **Bias**: model too simple → systematic error
 - **Variance**: model too complex → sensitive to noise
-  Balance needed for best generalization

Bias vs. Variance



Key Trade-offs



- Underfitting vs. Overfitting
 - Underfitting: can't capture signal (high bias)
 - Overfitting: memorizes training data (high variance)
-  Goal: fit patterns, not noise

Key Trade-offs



- Model simplicity vs. predictive power
 - Simple models: interpretable, fast
 - Complex models: powerful, harder to trust
-  Trade-off depends on **context**

Feature Engineering



- Transform **raw** data → **useful** signals 
- Domain knowledge as leverage 
- Scaling, encoding, dimensionality reduction 

Features → End-to-End Learning



- Hand-crafted features vs. automatic representation
- Neural networks = stacked nonlinearities
- End-to-end models reduce manual work

ML as Iteration



- Iterative loop is the heart of ML 
- Continuous experimentation mindset 
- Reproducibility = credibility + progress 

Takeaways



- Applied ML = **Data** + **Model** + **Evaluation** + **Iteration** 
- [Next] Workflow - scaling ML systematically 



Workflow

From Models to Workflows



- ML is more than training a model
- From prototype → product → lifecycle
- Workflows make ML real in practice
 - Applied ML is **ALWAYS** a team sport 🏈

The ML Lifecycle

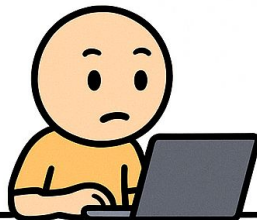


- Data → Train → Deploy → Monitor 
- Iterative and cyclical nature
- Feedback loop with users and environment 

Life as an OAI Member of Technical Staff



DATA



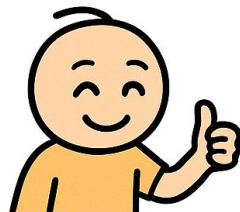
This time will be smooth.

TRAIN



Accuracy 95%!

DEPLOY



We did it!

MONITOR



Back to square one...



Data Pipeline

Data Collection



- Sources: logs, APIs, sensors, user input
- Bias and ethics in data collection
- Cost of free vs. expensive data

Data Cleaning & Preprocessing



- Handling missing values and outliers
- Scaling, normalization, encoding
- Importance of reproducibility in preprocessing

Feature Engineering Revisited



- Traditional feature crafting vs. learned features
- **Embeddings**, transfer learning
- Modern shift toward representation learning
 - What does it mean by  ?



**BREAK
TIME**





Model Deployment

Serving Models



- Batch vs. real-time inference
- REST APIs, microservices, cloud deployment
- Latency, scalability, cost trade-offs

Monitoring & Feedback



- Data drift and concept drift detection
- Performance monitoring in production
- User feedback as implicit supervision

Iterative Loop



- Train → Evaluate → Deploy → Monitor → Retrain
- Continuous integration and deployment (CI/CD for ML)
- Importance of fast iteration cycles

Takeaways



- **Workflow** = bridge from foundation to production
- Each step is **critical** to success of ML systems
- [Next] Production – scaling and sustaining ML



Production

From Workflow to Production



- Training a model is only the beginning
- Deployment brings new constraints and risks
- Models are part of larger socio-technical systems

Beyond Accuracy



- Latency: users expect instant results
- Cost: compute, storage, scaling
- Reliability & safety: uptime, robustness, compliance

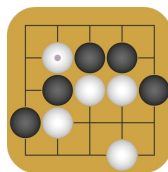


System Challenges

What Makes ML Systems Hard



- Black-box behavior, lack of clear specifications
- Outputs not always reproducible
- Hard to test exhaustively



Data & Scalability Issues



- ML learns patterns from data, not rules
- Inductive vs. deductive reasoning gap
- Scaling training and serving infrastructure

Failure Modes



- Overconfidence in wrong predictions
- Silent failures → hard to detect
- Cascading errors in pipelines

Monitoring in Production



- Detecting concept/data drift
- Live dashboards, anomaly alerts
- Collecting implicit and explicit feedback



People

Data Scientists vs. Software Engineers



- Scientists: accuracy, models, prototyping
- Engineers: cost, reliability, deployment
- Considerations: development speed vs. production stability

π -shaped People



- Broad range + deep expertise (in 1 or 2 areas)
- Example: engineer with ML + distributed systems
- Encourages team adaptability
- Critical thinking on AI suggestions

Cross-functional Teams



- Operators, product managers, designers
- Safety & security experts, lawyers, ethicists
- Collaboration essential for trust & adoption

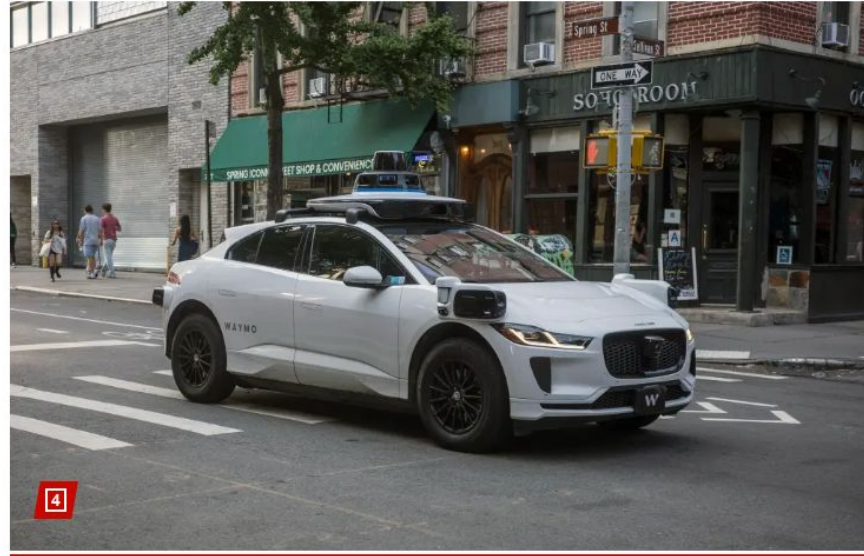


Case Study

NYC to test Waymo self-driving cars on crowded Manhattan and Brooklyn streets 08/25



The city will unleash the cars south of 112th Street in Manhattan in a program that a Waymo rep said Friday was already up and running.



Eight Waymo driverless cars will hit the road in Manhattan and Brooklyn.

Billy Becerra / NY Post

In Brooklyn, the driverless cars will roll out north of Atlantic Avenue and west of Carlton Street in neighborhoods such as Brooklyn Heights, Downtown Brooklyn and DUMBO.

Self-Driving Cars



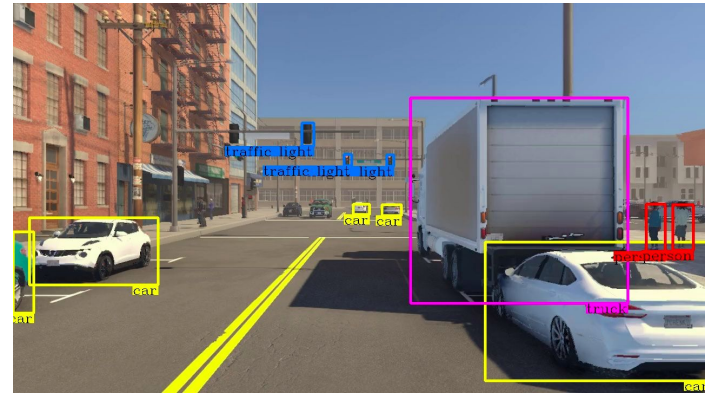
[Tasks] perception, mapping, planning, control



Rule-Based Approach



```
object = camera.get_object()  
if object.has_wheels():  
    if len(object.wheels) == 4: return "Car"  
    elif len(object.wheels) == 2: return "Bicycle"  
return "Unknown"
```



Supervised Learning



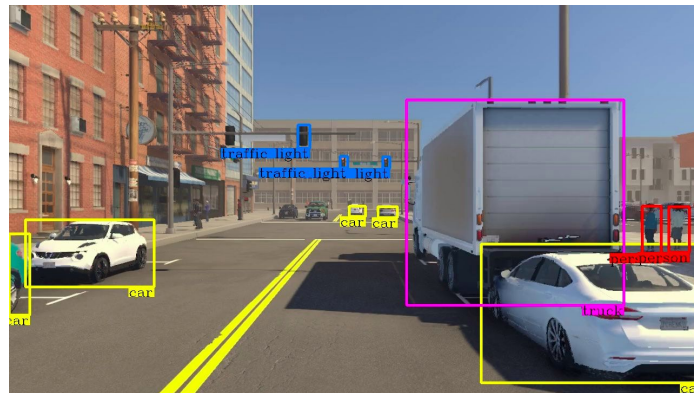
```
from sklearn.linear_model import LogisticRegression

# features: e.g., [num_wheels, has_engine]
X = [[4, 1], [2, 0], [3, 1]]
y = ["Car", "Bicycle", "Unknown"]

model = LogisticRegression().fit(X, y)

# predict
object = camera.get_object()
features = [len(object.wheels), int(object.has_engine())]
print(model.predict([features])[0])
```

- Learn boundaries between object categories
- Useful for object detection, lane marking recognition, and traffic sign classification



Reinforcement Learning



```
for episode in range(epochs):  
    state = env.reset()  
    while not done:  
        action = policy(state)  
        next_state, reward, done = env.step(action)  
        update_Q(state, action, reward, next_state)
```

- Agent interacts with environment
- Reward for safe lane keeping
- Penalties for collisions
- Used in planning & decision-making





Industrial insight: Why 99% Accuracy is a Disaster?

The Academic View

- Model Accuracy: 99% = "State of the Art" (A+ Grade)

The Waymo Reality

- The car makes decisions at 30 FPS (Frames Per Second)
- 99% Accuracy → 1% Error Rate
- 1% error every 100 frames
- 100 frames in 30 FPS $\approx$ 3.3 seconds

The Consequence

- Wait - Your 99% accurate model crashes the car every 3.3 seconds 😬

The Lesson:

- Applied ML is **NOT** about simply maximizing average accuracy
- It is about **suppressing the "Long Tail" of failures** - System Design POV

Summary



- [Foundation] ML = Data + Model + Evaluation + Iteration
- [Workflow] End-to-end lifecycle: Data → Train → Deploy → Monitor
- [Production] Beyond accuracy: reliability, business goal, scalability...
- [Application] Chatbot, self-driving car, starship...

Explore the ML Space

as vast as the stars and the sea

